

Discriminative Project

CNN / Deep Learning Based Automated Celebrity Face Detection System

Milestone 1

Group 01

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Git Repo Link:

<https://github.com/Yosephines/IE7615-Group1-Project1>

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Project proposal

Team: Yosephine Tong, Aditi Rao, Murat Akca, and Dario Garza

What we are building

We are building a system that identifies which of five celebrities appears in a photo. Later milestones will add the ability to find several faces in one image. We use Python with PyTorch and torchvision to build and train the models.

Our photos

We use CelebA identity IDs 7007, 2970, 2336, 7 and 4428. Each folder has enough photos to use 23 per person: 17 for learning, 3 for choosing the best model, and 3 for the final test. Using the same number of photos per person keeps the comparison balanced.

What we found

We compared three image-recognition models. ResNet18 identified the correct person in 10 of 15 test photos (66.7%). The deeper CNN got 6 correct (40.0%), and the small CNN got 5 correct (33.3%).

ResNet18 also performed best on the validation photos used to choose the model. We will use it in the next stages of the project. The trained models and everything needed to check these results are saved in the repository.

Team contributions

Member	Milestone 1 work
Yosephine Tong	Organized the repository, repeated training and prepared reports
Aditi Rao	Evaluated models, compared results and shared training logs
Murat Akca	Contribution details to be provided
Dario Garza	Prepared the initial data and model-training notebook

Next steps

Milestone 2 (end of Module 4): combine celebrity photos into new images and label each face with a box. Milestone 3 (mid-Module 5): train YOLOv8 to find those faces. Milestone 4 (end of Module 6): bring the system together and complete the final report.

Main risks

Our small dataset may not represent the photos the system will see later. Combined images may also look different from real scenes. We will keep training and test images separate and check the face labels carefully.

Classification results

Team: Yosephine Tong, Dario Garza and Aditi.

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Photos and preparation

We use five CelebA identity IDs: 7007, 2970, 2336, 7 and 4428. The folders contain 121 photos in total. We use 23 per person (115 total), leaving six unused. These are the dataset's ID labels; we have not verified celebrity names.

Photo set	Purpose	Total
Training	Teach the models	85
Validation	Choose the model	15
Test	Measure the final result	15

Each person has 17 training, 3 validation and 3 test photos. The file list is saved so every model uses the same photos. We found no identical image files across these groups.

We convert photos to color (RGB) and crop them to 224 x 224 pixels. During training, random crops and horizontal flips add variety. For validation and testing, we resize the shorter edge to 256 pixels and take a fixed center crop. Pixel values are adjusted using ImageNet's mean and standard deviation.

The three models

The small and deeper CNNs learn from our photos from the start. ResNet18 already learned useful image features from ImageNet, a large image dataset. We keep that part unchanged and train only its final layer to choose among our five identities.

How we trained them

Each model makes 30 passes through the training photos, called epochs. After each pass, we check it on the validation photos, which are used to choose the model. For each model, we save the version with the lowest validation loss (a measure of prediction error). We then choose the model with the most correct validation predictions and evaluate it on the separate test photos.

All models use Adam, a learning rate of 0.001 and batches of 16 images. The small CNN has two convolution layers (32 and 64 channels); the deeper CNN has four (32, 64, 128 and 256). Dropout rates are 0.25 and 0.35. The file shuffle uses seed 47 plus the identity number, so the same photos are selected each time.

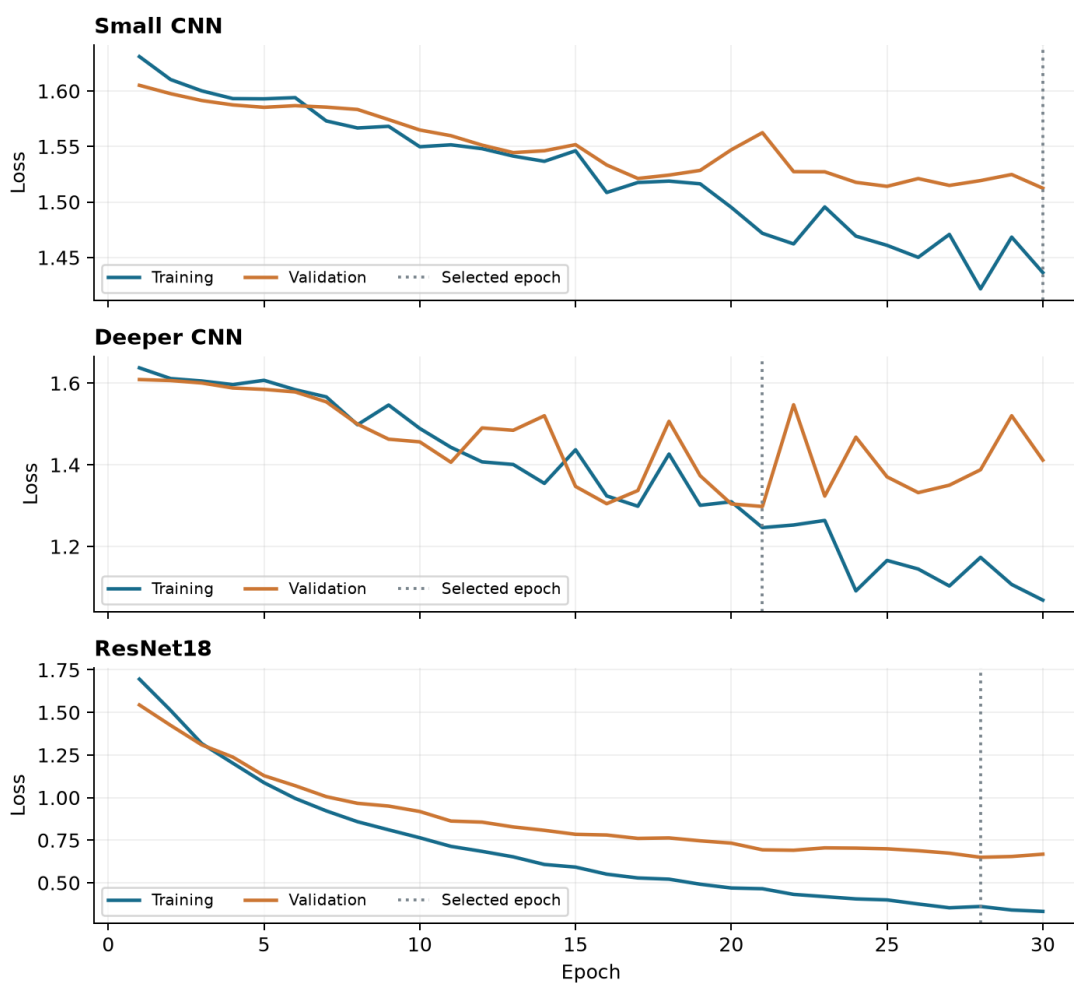
Results and training progress

Model	Validation	Test	Parameters (total / trained)	Training
Small CNN	26.7%	33.3% (5/15)	19,717 19,717	92.0 s
Deeper CNN	26.7%	40.0% (6/15)	389,701 389,701	146.0 s
ResNet18	80.0%	66.7% (10/15)	11,179,077 2,565	115.8 s

Parameters are the numbers stored inside a model. Training time includes validation and saving the best version. Measurements use an Intel Core i7-7700K CPU with two threads.

How prediction error changed during training

Loss measures prediction error; lower is better. Blue shows training photos and orange shows validation photos. The dotted line marks the saved version. These curves come directly from the training logs.

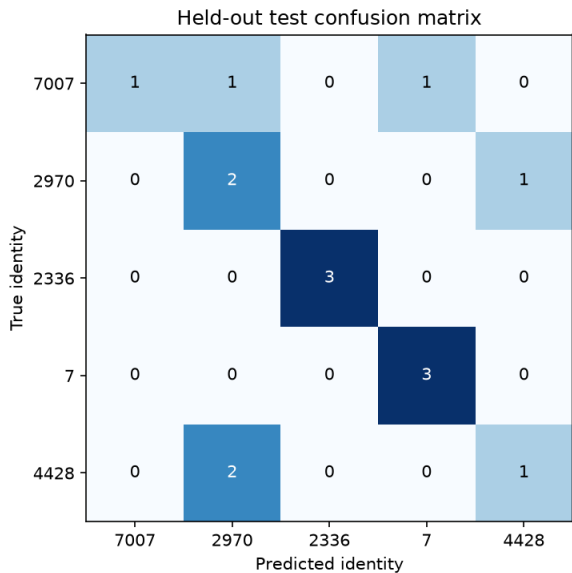


The small CNN improves slowly. The deeper CNN learns the training photos better, but its validation error rises and falls. ResNet18 has the lowest validation error. The selected versions are from training passes 30, 21 and 28 for the small CNN, deeper CNN and ResNet18.

Why we chose ResNet18

We chose ResNet18 because it correctly identified 12 of 15 validation photos (80.0%); each custom CNN identified 4 (26.7%). We made this choice before checking the test results. ResNet18's earlier training on ImageNet gives it useful image features when our own training set is small.

ResNet18 is larger and slower than the two custom CNNs. It contains about 11.18 million model values, called parameters, but we train only 2,565 in its final layer. It takes about 35.3 milliseconds to make a prediction on our CPU, excluding image preparation. We will use it as our starting model for later milestones because it performed best in this comparison.



This chart shows the test mistakes. Rows show the actual person; columns show the model's answer. Numbers on the diagonal are correct answers. For IDs 7007, 2970, 2336, 7 and 4428, the model gets 1, 2, 3, 3 and 1 of the three photos correct. It mistakes 4428 for 2970 twice.

Prediction time	Small CNN	Deeper CNN	ResNet18
Milliseconds per photo	11.5	18.1	35.3

We timed one photo at a time, after a warm-up, and report the median over five passes through the test set. Image loading and preparation are excluded.

What these results can tell us

The test set is small: just three photos per person. One correct or incorrect prediction changes the overall score by 6.7 percentage points. We need more varied photos to judge how well the model handles different lighting, poses and backgrounds.

Checking the results

The repository includes the photos, the list showing how they were divided, all three trained models, training logs and test predictions.

The README explains how to open notebooks 01-05, check the saved models and train them again. These results are saved as group1_repro_v1. Aditi's earlier results are kept separately for reference.